AWS CLOUD COMPUTING

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INTRODUCTION

Regardless of size, today's successful businesses all have online presences that need the use of data centres. The traditional data centre is on-premises, meaning that all its operations take place at a real location inside an office building for businesses. A data centre could mean that all computers under a desk, as large as an entire building loaded with blade servers in a climate-controlled room. It is managed by an internal IT team that is paid for and employed by the company that owns the data centre.

Imagine if, in addition to developing applications, you also need to establish a data centre for a firm when you want to make a straightforward website or digitize entire business operations. In order to run or maintain this data centre, you will need to purchase hardware (servers), software, and its licences, develop infrastructure, and a network, and recruit a team of professionals. You should determine how many servers you need before setting up the data centre because there is a potential that you will either over or under-provision hardware.

What if you could concentrate on creating applications without worrying about setting up your own data centre? Cloud computing enters the picture in this scenario.

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WHAT IS CLOUD COMPUTING?

Cloud computing refers to the distribution of computer services such as servers, data storage, databases, networking, software, analytics, and intelligence over the internet ("cloud") to provide flexible resources, quicker innovation, and cost savings. To put it another way, rather than investing in their own data centres, businesses can rent access to another party's infrastructure, such as storage, processing servers, and databases, from a cloud computing service provider and only pay for the resources they actually use.

You only pay for the cloud services that you really use, which lowers operational costs, improves infrastructure performance, and enables you to grow applications to meet changing business requirements.

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WHO USES CLOUD COMPUTING?

Building customer-facing online applications, data backup, delivering email and SMS notifications, virtual desktops, software development and testing, big data analytics, and disaster recovery are just a few of the many use cases that organisations of all shapes, sizes, and sectors are utilising the cloud for. As an illustration, telecom firms use cloud services to communicate with their clients by transmitting various kinds of messages. Companies that provide financial services are utilising the cloud to fuel real-time fraud prevention and detection.

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HOW DOES CLOUD COMPUTING WORK?

Let's separate cloud computing into two categories—front end and back end—to better understand how it functions. The client's computer or computer network makes up the front end. The cloud's back end is made up of multiple computers, servers, and data storage platforms. They are linked by a network, most frequently the Internet. The user's or client's side of the computer is called the front end. The system's "the cloud" portion serves as the back end.

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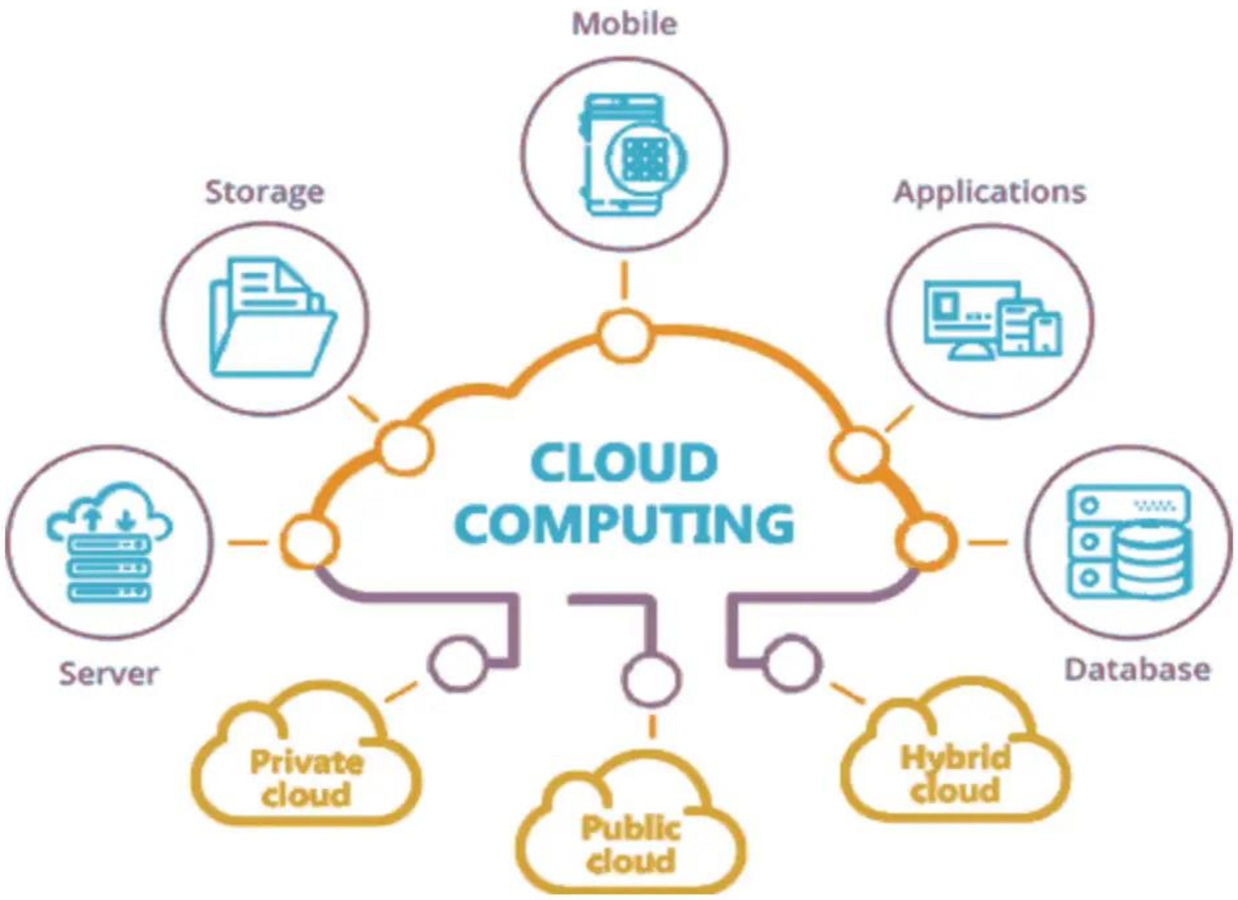
Cloud Computing



Cloud Computing

TYPES OF CLOUD SERVICES

Cloud Computing services are divided into three classes, according to the abstraction level of the capability provided and the service model of providers:

1. Infrastructure as a Service (LaaS in Cloud Computing)

2. Platform as a Service (PaaS in Cloud Computing)

3. Software as a Service (SaaS in Cloud Computing)

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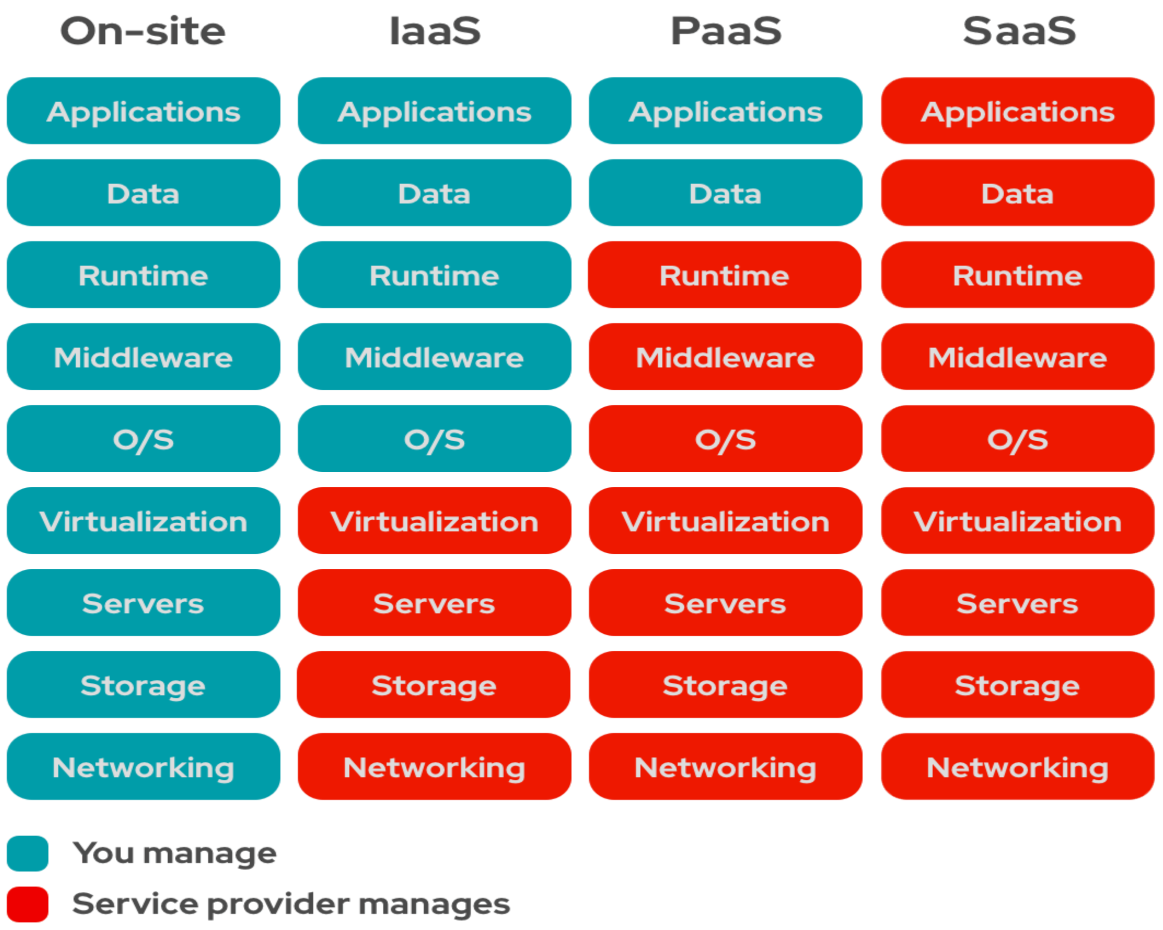

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BENEFITS OF CLOUD COMPUTING:

The benefits of cloud computing are numerous. Some of them are listed below:-

- Over the Internet, one can obtain applications used as utilities.
- Online application manipulation and configuration are always available.
- It is not necessary to install any software in order to use or access cloud applications.
- Through the PaaS model, cloud computing provides online tools for development and deployment as well as a runtime environment for programmes.
- Cloud resources are accessible through the network in a way that gives any kind of client platform-independent access.
- Self-service is available on demand using cloud computing.
- Without interacting with the cloud service provider, the resources can be used.
- Because it runs with maximum effectiveness and efficiency, cloud computing is very cost-effective.
- All that is needed is an Internet connection.
- Load balancing is a feature of cloud computing that increases its dependability.

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RISK RELATED TO CLOUD COMPUTING

Although there are many benefits of Cloud Computing, there are few disadvantages as well which you need to be aware of, such as:

- **Security & Privacy:** Although cloud service providers use the highest security measures and have earned industry certifications, putting sensitive information on outside service providers is always risky. Especially when it comes to handling sensitive data, security and privacy must be discussed in any discussion concerning data. You need to comprehend your cloud provider's shared responsibility model. You will still be responsible for everything that happens in your network and with your product.

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- **Vulnerability to Attack:** Every component in cloud computing is online, which exposes possible weaknesses. Even the finest teams occasionally experience serious attacks and security lapses.

- **Limited Control:** It is challenging for businesses to have an amount of control over cloud infrastructure since cloud services run on remote servers that are owned and managed by cloud service providers.

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AWS AVAILABILITY ZONES, REGIONS & EDGE LOCATIONS

Amazon Web Services (AWS) provides a global cloud computing infrastructure with a wide range of services, including computing, storage, and databases. To ensure high availability and low latency, AWS offers multiple geographic locations, including regions and Availability Zones (AZs).

- **Regions:** A region is a specific geographic location where AWS has multiple Availability Zones. There are currently 24 AWS regions globally, each region contains multiple data centers.
- **Availability Zones (AZs):** An Availability Zone is a physically separate data center within an AWS region. Each Availability Zone is isolated, but connected to other Availability Zones in the same region, offering low latency and high throughput network connections. By spreading your application across multiple Availability Zones, you can build high availability into your applications.

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- **Edge Locations:** An edge location is a point of presence (PoP) in a specific geographic location that is close to end-users. AWS uses edge locations to cache content from Amazon CloudFront, a web service for content delivery. The purpose of edge locations is to reduce the latency and improve the performance of web applications and content delivery by serving content closer to end-users.

By having multiple regions, Availability Zones, and edge locations, AWS provides customers with the ability to run their applications in a highly available and scalable manner, with low latency and high performance.

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DIFFERENCE BETWEEN AWS, AZURE & GCP

AWS, Azure, and GCP are the three leading cloud computing platforms, each with its own strengths and offerings.

- **Amazon Web Services (AWS)** is a collection of remote computing services (also called web services) that make up a cloud computing platform, offered by Amazon.com. It provides a wide range of services, including computing, storage, and databases.
- **Microsoft Azure** is a cloud computing platform and infrastructure created by Microsoft for building, deploying, and managing applications and services through a global network of Microsoft-managed data centers. It offers a range of services, including virtual machines, storage, and databases.

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Google Cloud Platform (GCP) is a cloud computing platform and infrastructure created by Google for building, deploying, and managing applications and services on the same infrastructure that Google uses internally for end-user products like Google Search and YouTube. It provides a range of services, including computing, storage, and databases, as well as machine learning and big data analytics.

In general, AWS is considered the most mature and feature-rich of the three, while Azure is known for its strong support for Windows and .NET, and GCP is known for its strong support for machine learning and big data analytics. The choice of platform ultimately depends on the specific needs and requirements of the user.

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Cloud Comparison
Azure vs. AWS vs. Google Compute

	Azure	aws	
Available Regions	Azure Regions	AWS Regions and Zones	Google Compute Regions & Zones
Compute Services	Virtual Machines	Elastic Compute Cloud (EC2)	Compute Engine
App Hosting	Azure Cloud Services	Amazon Elastic Beanstalk	Google App Engine
Serverless Computing	Azure Functions	AWS Lambda	Google Cloud Functions
Container Support	Azure Container Service	EC2 Container Service	Container Engine
Scaling Options	Azure Autoscale	Auto Scaling	Autoscaler
Object Storage	Azure Blob Storage	Amazon Simple Storage (S3)	Cloud Storage
Block Storage	Azure Managed Storage	Amazon Elastic Block Storage	Persistent Disk
Content Delivery Network (CDN)	Azure CDN	Amazon CloudFront	Cloud CDN
SQL Database Options	Azure SQL Database	Amazon RDS	Cloud SQL
NoSQL Database Options	Azure DocumentDB	AWS DynamoDB	Cloud Datastore
Virtual Network	Azure Virtual Network	Amazon VPC	Cloud Virtual Network
Private Connectivity	Azure Express Route	AWS Direct Connect	Cloud Interconnect
DNS Services	Azure Traffic Manager	Amazon Route 53	Cloud DNS
Log Monitoring	Azure Operational Insights	Amazon CloudTrail	Cloud Logging
Performance Monitoring	Azure Application Insights	Amazon CloudWatch	Stackdriver Monitoring
Administration and Security	Azure Active Directory	AWS Identity and Access Management (IAM)	Cloud Identity and Access Management (IAM)
Compliance	Azure Trust Center	AWS CloudHSM	Google Cloud Platform Security
Analytics	Azure Stream Analytics	Amazon Kinesis	Cloud Dataflow
Automation	Azure Automation	AWS Opsworks	Compute Engine Management
Management Services & Options	Azure Resource Manager	Amazon Cloudformation	Cloud Deployment Manager
Notifications	Azure Notifcation Hub	Amazon Simple Notification Service (SNS)	None
Load Balancing	Load Balancing for Azure	Elastic Load Balancing	Cloud Load Balancing

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AMAZON EC2

Amazon Elastic Compute Cloud (Amazon EC2) is a cloud computing service provided by Amazon Web Services (AWS) that provides scalable computing capacity in the cloud. EC2 enables users to rent virtual computers on which to run their own computer applications. With EC2, users can easily launch and configure virtual machines (known as "instances") with a variety of operating systems and configurations, and then connect to these instances to run applications and store data.

EC2 provides a range of benefits, including the ability to scale computing resources up or down as needed, the ability to launch instances in multiple availability zones for high availability, and the ability to use Amazon Machine Images (AMIs) to quickly launch pre-configured instances with common software stacks. EC2 also integrates with other AWS services, such as Amazon S3 for storage, Amazon RDS for database services, and Amazon VPC for networking and security.

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BENEFITS OF EC2

Amazon Elastic Compute Cloud (Amazon EC2) provides several benefits to users, including:

- **Scalability:** EC2 enables users to quickly scale computing resources up or down as needed to accommodate changes in demand, making it easy to handle spikes in traffic or temporary increases in computing needs.
- **Flexibility:** EC2 provides a wide range of instance types and configurations, making it easy to choose the right instance type for a specific use case, such as compute-intensive applications, memory-intensive applications, or storage-intensive applications.
- **Cost-effectiveness:** EC2 provides a cost-effective way to rent computing resources on an as-needed basis, without having to invest in expensive hardware. Additionally, EC2 provides several cost-saving features, such as Amazon EC2 Spot Instances, which allow users to bid on unused EC2 capacity at a discount.
- **High availability:** EC2 instances can be launched in multiple availability zones, providing users with the ability to build highly available applications that can survive failures in a single location.

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- **Integration with other AWS services:** EC2 integrates with a wide range of other AWS services, including Amazon S3 for storage, Amazon RDS for database services, and Amazon VPC for networking and security, making it easy to build complete cloud-based solutions.

- **Security:** EC2 provides a secure environment for running applications and storing data, with features such as security groups and key pairs to control access to instances and data, and encryption options for data at rest and in transit.

- **Easy management:** EC2 provides a simple and intuitive management interface, making it easy to launch, configure, and manage instances, and monitor resource utilization and performance.

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FEATURES OF AMAZON EC2:

- **Virtual Machine Instances:** EC2 provides a wide range of virtual machine instances, including instances optimized for compute, memory, and storage, as well as instances with GPU acceleration for specialized workloads.
- **Auto Scaling:** EC2 enables users to automatically scale computing resources up or down based on demand, making it easy to handle spikes in traffic and fluctuations in computing needs.
- **Load Balancing:** EC2 integrates with Amazon Elastic Load Balancer to provide highly available and scalable applications.
- **Elastic Block Store (EBS):** EC2 integrates with Amazon EBS to provide persistent block-level storage for EC2 instances.
- **Amazon Machine Images (AMIs):** EC2 provides a library of Amazon Machine Images (AMIs), which are pre-configured virtual machine images that can be used to quickly launch instances with common software stacks.

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- **Networking and Security:** EC2 provides a virtual private cloud (VPC) for isolating instances and controlling network access, as well as security groups and key pairs for controlling access to instances.

- **Monitoring and Management:** EC2 provides tools for monitoring resource utilization and performance, as well as for managing instances and automating common tasks.

- **Cost Optimization:** EC2 provides cost-saving features, such as Amazon EC2 Spot Instances and Amazon EC2 Reserved Instances, which allow users to bid on unused EC2 capacity or reserve capacity in advance at a discount.

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EC2 CREATION STEPS

- **Log in to the AWS Management Console:** Go to the AWS Management Console and sign in with your AWS account.
- **Launch an EC2 instance:** In the EC2 Dashboard, click the "Launch Instance" button. Select an Amazon Machine Image (AMI) that matches your desired operating system and software stack. Choose an instance type that matches your computing needs.
- **Configure instance details:** Select the number of instances you want to launch and configure any additional options, such as network settings, storage options, and security groups.
- **Add storage:** Configure the instance's storage by attaching Amazon Elastic Block Store (EBS) volumes or adding instance storage.

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- **Configure security**: Set up security groups and key pairs to control access to the instance.
  Launch the instance: Review the instance details and launch the instance.

- **Connect to the instance**: Connect to the instance using the public DNS name or IP address, along with the private key file associated with the key pair you specified during the instance launch process.

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